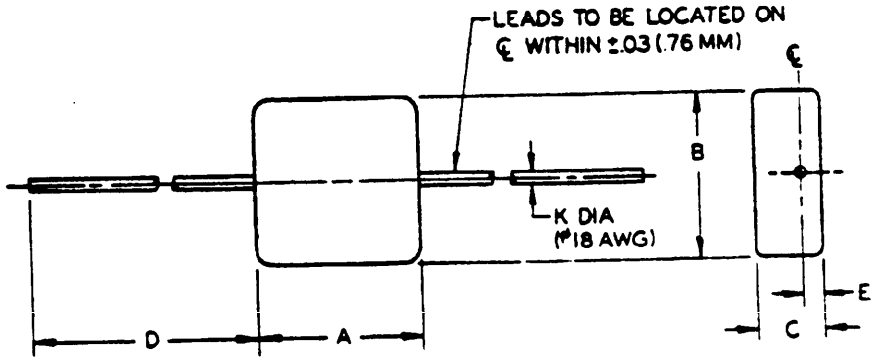
MIL-C-5/4E
20 September 1966
SUPERSEDING
MIL-C-5/4D
1~~5~~ November 1963

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,

~~STYLE~~ CM30

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.



DIMENSIONS						
DIM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.77	.80	.86	19.56	20.32	21.84
B	.77	.80	.86	19.56	20.32	21.84
C	.23	.25	.28	5.84	6.35	7.11
D	1.12	---	---	28.45	---	---
E	.08	---	---	2.03	---	---
K	.038	.040	.042	.97	1.02	1.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight .17 ounce.
4. Unless otherwise specified, tolerance is .xx ±.02.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Symbol	Capacitance tolerance
F	±1 percent
G	±2 percent
J	±5 percent

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55°C to +85°C
O	-55°C to +125°C
P	-55°C to +150°C

Vibration grade: 3 (10 to 2,000 cps).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
C	-200 to +200	±(0.5 percent + 0.1 pf)
E	-20 to +100	±(0.1 percent + 0.1 pf)
F	0 to +70	±(0.05 percent + 0.1 pf)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202: Condition D (100,000 ft).
Test potential - 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 85°C:
1100 to 5000 pf - 15,000 megohms, minimum
5100 to 7500 pf - 75 megohm microfarads, min.
At 125°C:
10,000 megohms, minimum
At 150°C:
0 to 1500 pf - 5000 megohms, minimum
1600 to 7500 pf - 7.5 megohm microfarads, min.

Dissipation factor, maximum: See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Vibration:
Method 204 of MIL-STD-202, condition D.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be as follows:
Symbol O - +125°C.
Symbol P - +150°C.

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Solderability:
Method 208 of MIL-STD-202.

Terminal strength:
Pull test - Capacitor held by one terminal and 5 pound load gradually applied to other terminal and maintained for at least 5 seconds.
Twist test - 5 alternate 360° rotations.

Moisture resistance:
Method 106 of MIL-STD-202.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at $-55\ ^{+0}_{-2}$°C, immediately followed by qualification or performance test, as applicable.
Qualification test - 2,000 hours at applicable high test temperature with 150 percent of rated dc working voltage applied.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.
Performance check - Conditions and requirements in accordance with qualification test except test duration shall be 250 hours.
Continuation test - Units that have completed the 250 hour performance shall be continued for an additional 1750 hours and meet the requirements for the qualification test.

Qualification inspection: Qualification inspection (Group A) is required.

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TABLE I. Style CM30.

Type designation 1/	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available for capacitance tolerances		
				±1% (F)	±2% (G)	±5% (J)
CM30-D112--3	1100	500	N, O, P	CEF	CEF	CEF
CM30-D122--3	1200	500	N, O, P	CEF	CEF	CEF
CM30-D132--3	1300	500	N, O, P	CEF	CEF	CEF
CM30-D152--3	1500	500	N, O, P	CEF	CEF	CEF
CM30-D162--3	1600	500	N, O, P	CEF	CEF	CEF
CM30-D182--3	1800	500	N, O, P	CEF	CEF	CEF
CM30-D202--3	2000	500	N, O, P	CEF	CEF	CEF
CM30-D222--3	2200	500	N, O, P	CEF	CEF	CEF
CM30-D242--3	2400	500	N, O, P	CEF	CEF	CEF
CM30-D272--3	2700	500	N, O, P	CEF	CEF	CEF
CM30-D302--3	3000	500	N, O, P	CEF	CEF	CEF
CM30-D332--3	3300	500	N, O, P	CEF	CEF	CEF
CM30-D362--3	3600	500	N, O	CEF	CEF	CEF
CM30-D392--3	3900	500	N, O	CEF	CEF	CEF
CM30-D432--3	4300	500	N, O	CEF	CEF	CEF
CM30-D472--3	4700	500	N, O	CEF	CEF	CEF
CM30-D512--3	5100	500	N, O	CEF	CEF	CEF
CM30-D562--3	5600	500	N, O	CEF	CEF	CEF
CM30-D622--3	6200	500	N, O	CEF	CEF	CEF
CM30-D682--3	6800	500	N, O	CEF	CEF	CEF
CM30-D752--3	7500	500	N, O	CEF	CEF	CEF

1/ Complete type designation will include additional symbols to indicate characteristic, capacitance tolerance and temperature range.

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Supersession data: Capacitors with voltage ratings of 1,000 volts and greater, formerly covered by this specification sheet, with part numbers M5/04-0361 thru -0639, are no longer authorized. Since there has never been a qualified source for these items, no replacement data is offered. For capacitors with voltage ratings of 500 volts and less, see table II for replacement data.

TABLE II. Part number to replacement type designation.

Dash No.	Replacement item	Dash No.	Replacement item	Dash No.	Replacement item
0001	CM30CD112FN3	0057	CM30FD122FP3	0114	CM30ED152FP3
0002	CM30CD112GN3	0058	CM30FD122GP3	0115	CM30ED152GP3
0003	CM30CD112JN3	0059	CM30FD122JP3	0116	CM30ED152JP3
0004	CM30ED112FN3	0061	CM30CD132FN3	0117	CM30FD152FP3
0005	CM30ED112GN3	0062	CM30CD132GN3	0118	CM30FD152GP3
0006	CM30ED112JN3	0063	CM30CD132JN3	0119	CM30FD152JP3
0007	CM30FD112FN3	0064	CM30ED132FN3	0121	CM30CD162FN3
0008	CM30FD112GN3	0065	CM30ED132GN3	0122	CM30CD162GN3
0009	CM30FD112JN3	0066	CM30ED132JN3	0123	CM30CD162JN3
0011	CM30CD112FO3	0067	CM30FD132FN3	0124	CM30ED162FN3
0012	CM30CD112GO3	0068	CM30FD132GN3	0125	CM30ED162GN3
0013	CM30CD112JO3	0069	CM30FD132JN3	0126	CM30ED162JN3
0014	CM30ED112FO3	0071	CM30CD132FO3	0127	CM30FD162FN3
0015	CM30ED112GO3	0072	CM30CD132GO3	0128	CM30FD162GN3
0016	CM30ED112JO3	0073	CM30CD132JO3	0129	CM30FD162JN3
0017	CM30FD112FO3	0074	CM30ED132FO3	0131	CM30CD162FO3
0018	CM30FD112GO3	0075	CM30ED132GO3	0132	CM30CD162GO3
0019	CM30FD112JO3	0076	CM30ED132JO3	0133	CM30CD162JO3
0021	CM30CD112FP3	0077	CM30FD132FO3	0134	CM30ED162FO3
0022	CM30CD112GP3	0078	CM30FD132GO3	0135	CM30ED162GO3
0023	CM30CD112JP3	0079	CM30FD132JO3	0136	CM30ED162JO3
0024	CM30ED112FP3	0081	CM30CD132FP3	0137	CM30FD162FO3
0025	CM30ED112GP3	0082	CM30CD132GP3	0138	CM30FD162GO3
0026	CM30ED112JP3	0083	CM30CD132JP3	0139	CM30FD162JO3
0027	CM30FD112FP3	0084	CM30ED132FP3	0141	CM30CD162FP3
0028	CM30FD112GP3	0085	CM30ED132GP3	0142	CM30CD162GP3
0029	CM30FD112JP3	0086	CM30ED132JP3	0143	CM30CD162JP3
0031	CM30CD122FN3	0087	CM30FD132FP3	0144	CM30ED162FP3
0032	CM30CD122GN3	0088	CM30FD132GP3	0145	CM30ED162GP3
0033	CM30CD122JN3	0089	CM30FD132JP3	0146	CM30ED162JP3
0034	CM30ED122FN3	0091	CM30CD152FN3	0147	CM30FD162FP3
0035	CM30ED122GN3	0092	CM30CD152GN3	0148	CM30FD162GP3
0036	CM30ED122JN3	0093	CM30CD152JN3	0149	CM30FD162JP3
0037	CM30FD122FN3	0094	CM30ED152FN3	0151	CM30CD182FN3
0038	CM30FD122GN3	0095	CM30ED152GN3	0152	CM30CD182GN3
0039	CM30FD122JN3	0096	CM30ED152JN3	0153	CM30CD182JN3
0041	CM30CD122FO3	0097	CM30FD152FN3	0154	CM30ED182FN3
0042	CM30CD122GO3	0098	CM30FD152GN3	0155	CM30ED182GN3
0043	CM30CD122JO3	0099	CM30FD152FN3	0156	CM30ED182JN3
0044	CM30ED122FO3	0101	CM30CD152FO3	0157	CM30FD182FN3
0045	CM30ED122GO3	0102	CM30CD152GO3	0158	CM30FD182GN3
0046	CM30ED122JO3	0103	CM30CD152JO3	0159	CM30FD182JN3
0047	CM30FD122FO3	0104	CM30ED152FO3	0161	CM30CD182FO3
0048	CM30FD122GO3	0105	CM30ED152GO3	0162	CM30CD182GO3
0049	CM30FD122JO3	0106	CM30ED152JO3	0163	CM30CD182JO3
0051	CM30CD122FP3	0107	CM30FD152FO3	0164	CM30ED182FO3
0052	CM30CD122GP3	0108	CM30FD152GO3	0165	CM30ED182GO3
0053	CM30CD122JP3	0109	CM30FD152JO3	0166	CM30ED182JO3
0054	CM30ED122FP3	0111	CM30CD152FP3	0167	CM30FD182FO3
0055	CM30ED122GP3	0112	CM30CD152GP3	0168	CM30FD182GO3
0056	CM30ED122JP3	0113	CM30CD152JP3	0169	CM30FD182JO3

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TABLE II. Part number to replacement type designation. - Continued

Dash No.	Replacement item	Dash No.	Replacement item	Dash No.	Replacement item
0171	CM30CD182FP3	0231	CM30CD222FP3	0291	CM30CD272FP3
0172	CM30CD182GP3	0232	CM30CD222GP3	0292	CM30CD272GP3
0173	CM30CD182JP3	0233	CM30CD222JP3	0293	CM30CD272JP3
0174	CM30ED182FP3	0234	CM30ED222FP3	0294	CM30ED272FP3
0175	CM30ED182GP3	0235	CM30ED222GP3	0295	CM30ED272GP3
0176	CM30ED182JP3	0236	CM30ED222JP3	0296	CM30ED272JP3
0177	CM30FD182FP3	0237	CM30FD222FP3	0297	CM30FD272FP3
0178	CM30FD182GP3	0238	CM30FD222GP3	0298	CM30FD272GP3
0179	CM30FD182JP3	0239	CM30FD222JP3	0299	CM30FD272JP3
0181	CM30CD202FN3	0241	CM30CD242FN3	0301	CM30CD302FN3
0182	CM30CD202GN3	0242	CM30CD242GN3	0302	CM30CD302GN3
0183	CM30CD202JN3	0243	CM30CD242JN3	0303	CM30CD302JN3
0184	CM30ED202FN3	0244	CM30ED242FN3	0304	CM30ED302FN3
0185	CM30ED202GN3	0245	CM30ED242GN3	0305	CM30ED302GN3
0186	CM30ED202JN3	0246	CM30ED242JN3	0306	CM30ED302JN3
0187	CM30FD202FN3	0247	CM30FD242FN3	0307	CM30FD302FN3
0188	CM30FD202GN3	0248	CM30FD242GN3	0308	CM30FD302GN3
0189	CM30FD202JN3	0249	CM30FD242JN3	0309	CM30FD302JN3
0191	CM30CD202FO3	0251	CM30CD242FO3	0311	CM30CD302FO3
0192	CM30CD202GO3	0252	CM30CD242GO3	0312	CM30CD302GO3
0193	CM30CD202JO3	0253	CM30CD242JO3	0313	CM30CD302JO3
0194	CM30ED202FO3	0254	CM30ED242FO3	0314	CM30ED302FO3
0195	CM30ED202GO3	0255	CM30ED242GO3	0315	CM30ED302GO3
0196	CM30ED202JO3	0256	CM30ED242JO3	0316	CM30ED302JO3
0197	CM30FD202FO3	0257	CM30FD242FO3	0317	CM30FD302FO3
0198	CM30FD202GO3	0258	CM30FD242GO3	0318	CM30FD302GO3
0199	CM30FD202JO3	0259	CM30FD242JO3	0319	CM30FD302JO3
0201	CM30CD202FP3	0261	CM30CD242FP3	0321	CM30CD302FP3
0202	CM30CD202GP3	0262	CM30CD242GP3	0322	CM30CD302GP3
0203	CM30CD202JP3	0263	CM30CD242JP3	0323	CM30CD302JP3
0204	CM30ED202FP3	0264	CM30ED242FP3	0324	CM30ED302FP3
0205	CM30ED202GP3	0265	CM30ED242GP3	0325	CM30ED302GP3
0206	CM30ED202JP3	0266	CM30ED242JP3	0326	CM30ED302JP3
0207	CM30FD202FP3	0267	CM30FD242FP3	0327	CM30FD302FP3
0208	CM30FD202GP3	0268	CM30FD242GP3	0328	CM30FD302GP3
0209	CM30FD202JP3	0269	CM30FD242JP3	0329	CM30FD302JP3
0211	CM30CD222FN3	0271	CM30CD272FN3	0331	CM30CD332FN3
0212	CM30CD222GN3	0272	CM30CD272GN3	0332	CM30CD332GN3
0213	CM30CD222JN3	0273	CM30CD272JN3	0333	CM30CD332JN3
0214	CM30ED222FN3	0274	CM30ED272FN3	0334	CM30ED332FN3
0215	CM30ED222GN3	0275	CM30ED272GN3	0335	CM30ED332GN3
0216	CM30ED222JN3	0276	CM30ED272JN3	0336	CM30ED332JN3
0217	CM30FD222FN3	0277	CM30FD272FN3	0337	CM30FD332FN3
0218	CM30FD222GN3	0278	CM30FD272GN3	0338	CM30FD332GN3
0219	CM30FD222JN3	0279	CM30FD272JN3	0339	CM30FD332JN3
0221	CM30CD222FO3	0281	CM30CD272FO3	0341	CM30CD332FO3
0222	CM30CD222GO3	0282	CM30CD272GO3	0342	CM30CD332GO3
0223	CM30CD222JO3	0283	CM30CD272JO3	0343	CM30CD332JO3
0224	CM30ED222FO3	0284	CM30ED272FO3	0344	CM30ED332FO3
0225	CM30ED222GO3	0285	CM30ED272GO3	0345	CM30ED332GO3
0226	CM30ED222JO3	0286	CM30ED272JO3	0346	CM30ED332JO3
0227	CM30FD222FO3	0287	CM30FD272FO3	0347	CM30FD332FO3
0228	CM30FD222GO3	0288	CM30FD272GO3	0348	CM30FD332GO3
0229	CM30FD222JO3	0289	CM30FD272JO3	0349	CM30FD332JO3

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TABLE II. Part number to replacement type designation. - Continued

Dash No.	Replacement item	Dash No.	Replacement item	Dash No.	Replacement item
0351	CM30CD332FP3	0354	CM30ED332FP3	0357	CM30FD332FP3
0352	CM30CD332GP3	0355	CM30ED332GP3	0358	CM30FD332GP3
0353	CM30CD332JP3	0356	CM30ED332JP3	0359	CM30FD332JP3

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH, AS, OS, EC
Air Force - 11, 17, 85

User activities:
Army -
Navy - MC, CG
Air Force - 14, 19

Preparing activity:
Navy - SH

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